



NOTRE DAME UNIVERSITY BANGLADESH

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IoT-Based Smart Farm Threat Monitoring System Using Cisco Packet Tracer

COURSE: COMPUTER NETWORK LAB

COURSE CODE: CSE-3204

Submitted By:

Maria Fardus

Nazifa Tasnim

Syeda Nazme Rahman

Raisha Anjum

Nadia Afroj

ID: 069223005101003

ID: 069223005101007

ID: 069223005101011

ID: 069223005101019

ID: 069223005101023

Submitted To:

Dr.Fernaz Narin Nur

Adjunct Professor

Department of CSE, NDUB

Date: 16-06-2025

Contents

Abstract	2
1 Introduction	2
2 Technical Description	2
3 Methodology	3
3.1 System Modules	3
3.2 DHCP Integration	4
3.3 Flowchart	4
4 Connectivity	7
4.1 Home Gateway and DHCP	7
4.2 Ping Test	7
5 Analysis and Discussion	7
6 Advantages & Disadvantages	9
6.1 Advantages of IoT Devices in This Project	9
6.2 Advantages	9
6.3 Disadvantages	10
7 Software Implementation	11
8 Further Work	11
9 Conclusion	12
Conflicts of Interest	12

Abstract

This paper presents an IoT-based Smart Farm Threat Monitoring System designed to enhance agricultural security, efficiency, and sustainability. The system integrates multiple subsystems including wild animal intrusion detection, air quality monitoring, smart lighting, automated door barricades, soil moisture sensing, solar energy harvesting, fire detection, and temperature/humidity monitoring. Each IoT device is connected through a centralized network with IP addressing dynamically handled via a configured DHCP server in Cisco Packet Tracer. The smart farm setup enables real-time automation and remote control via smart devices, transforming traditional farming into an intelligent, self-regulating environment.

Subject Areas

IoT Systems, Smart Agriculture, Environmental Monitoring, Sensor Networks, Wireless Communication

Keywords

Smart Farm, IoT, Wild Animal Detection, Soil Moisture, Fire Sensor, Solar Panel, Humidity, DHCP, Cisco Packet Tracer

1 Introduction

Smart farming represents the integration of information and communication technologies into agriculture. Traditional farming is often labor-intensive and inefficient in terms of threat monitoring and environmental control. With IoT and automation, a farm can become self-monitoring and adaptive. This project simulates such a system using Cisco Packet Tracer, where sensors and actuators are controlled remotely and configured to operate based on real-time environmental conditions. DHCP is used for automatic IP configuration of devices, simplifying network management and improving scalability.

2 Technical Description

All IoT devices are wirelessly connected to a central Home Gateway. Instead of assigning static IPs manually, a DHCP server is configured to dynamically assign IP addresses to hosts and IoT components. DHCP (Dynamic Host Configuration Protocol) allows devices to receive IP addresses automatically within a predefined range, reducing the risk of IP conflicts and simplifying administration.

The DHCP server is configured with:

IP Pool Range: e.g., 192.168.1.100 – 192.168.1.200

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Each device requests an IP from the server upon boot or reconnection.

3 Methodology

3.1 System Modules

The IoT Smart Farm Threat Monitoring System uses multiple automated modules to ensure farm safety, optimize resource use, and monitor threats. Each subsystem relies on sensors and microcontroller logic to perform specific tasks, working together to maintain a secure and efficient farming environment.

- **Wild Animal Intrusion Detection:** Detects movement of wild animals using a PIR sensor and triggers a siren to scare them away and alert the farmer.
 - PIR sensor monitors movement near farm boundaries.
 - Activates siren on detecting unauthorized animals.
- **Air Quality Monitoring:** Monitors CO₂ levels in enclosed areas and turns on exhaust fans when levels exceed safe limits to maintain air quality.
 - CO₂ sensor tracks gas concentration.
 - Exhaust fans activate automatically if CO₂ is high.
- **Smart Lighting System:** Adjusts farm lighting based on ambient light using a light sensor to save energy.
 - Light sensor measures natural light intensity.
 - Lights switch on/off accordingly.
- **Authorized Door Barricade System:** Controls access with RFID cards, opening the door only for authorized personnel.
 - RFID reader scans ID cards.
 - Microcontroller verifies and grants access.
- **Soil Moisture Detection:** Monitors soil moisture and controls sprinklers to water crops only when needed.
 - Soil moisture sensor detects water levels.
 - Sprinklers activate when moisture is low.
- **Solar Panel Energy System:** Manages solar energy, powering IoT devices when battery charge is above 30%.
 - Solar panels charge batteries.
 - Microcontroller distributes power to devices.
- **Fire Detection System:** Detects smoke and activates sprinklers and alarms to respond to fire threats immediately.
 - Smoke detector senses fire signs.
 - Sprinklers and sirens activate on detection.

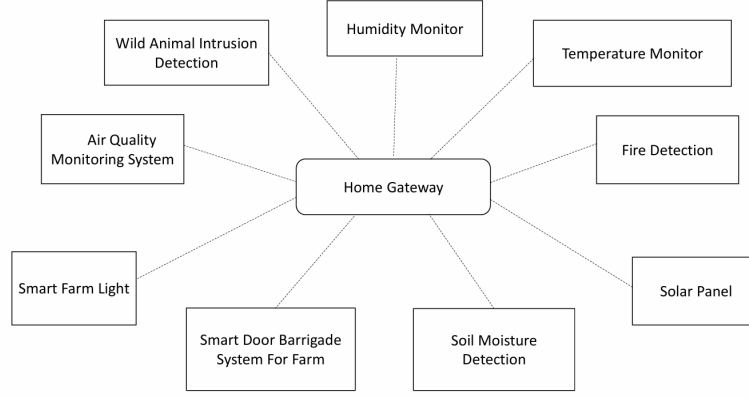


Figure 1: Block diagram of the design.

3.2 DHCP Integration

Instead of assigning IPs manually: A DHCP server is configured in the network (either standalone or inside the home gateway).

Devices send a DHCP Discover message upon connection.

The DHCP server responds with an Offer including:

Available IP address

Subnet mask

Default gateway

The device sends a DHCP request to accept the offer.

DHCP server confirms with an Acknowledgement.

This process ensures that all devices in the farm are uniquely and correctly addressed without manual intervention.

3.3 Flowchart

The below flowchart procedure is used for all the systems active in this design. First, the sensors sense their respective parameters (according to the pre-set conditions) and the values sensed are converted into the appropriate formats

which are then fed to the actuators. The actuators are activated based on the input given following predefined conditions as shown in Figure 6 below.



Figure 2: Connected IoT devices

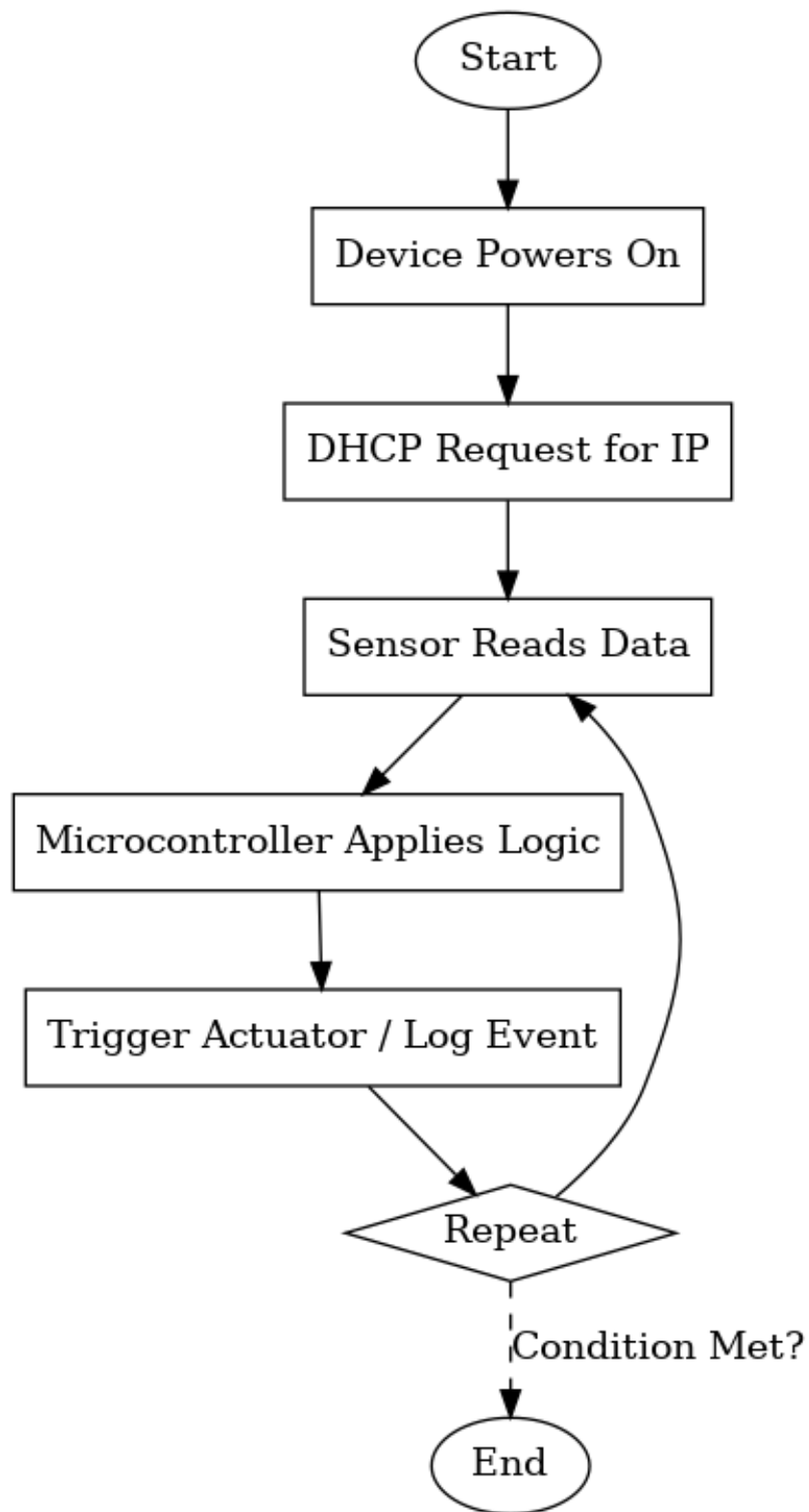


Figure 3: General flow chart of the design methodology.

4 Connectivity

4.1 Home Gateway and DHCP

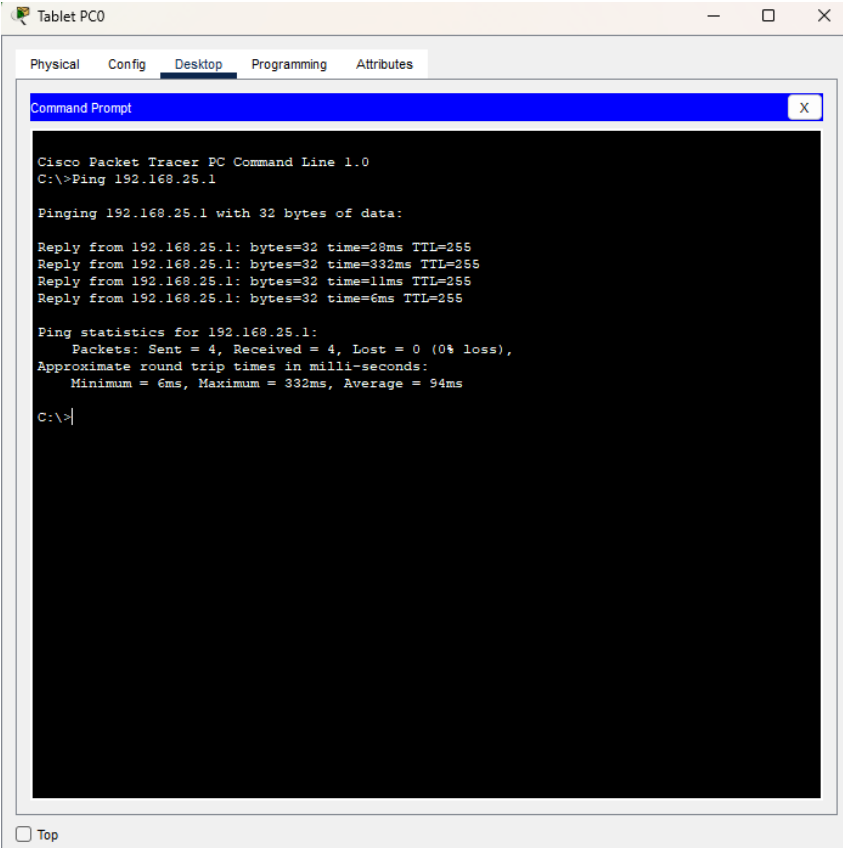
SSID: smart_farm, Gateway IP: 192.168.1.1... DHCP range configured in the server: 192.168.1.100 - 192.168.1.200

Devices obtain IPs automatically, eliminating manual errors.

All devices on the same network via Wi-Fi connection to the gateway.

4.2 Ping Test

Ping tests verify connectivity. Since DHCP is used, device IPs are checked via device information tables or the command prompt (ipconfig) and successfully pinged to confirm network integrity.



```
Tablet PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>Ping 192.168.25.1

Pinging 192.168.25.1 with 32 bytes of data:

Reply from 192.168.25.1: bytes=32 time=20ms TTL=255
Reply from 192.168.25.1: bytes=32 time=332ms TTL=255
Reply from 192.168.25.1: bytes=32 time=11ms TTL=255
Reply from 192.168.25.1: bytes=32 time=6ms TTL=255

Ping statistics for 192.168.25.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 332ms, Average = 94ms

C:\>|
```

Figure 4: Ping results

5 Analysis and Discussion

The simulation of the IoT-based Smart Farm Threat Monitoring System in Cisco Packet Tracer demonstrated the successful integration and functionality of all modules. Devices across the farm were able to connect to the DHCP-enabled network and obtain IP

addresses dynamically, which ensured efficient communication without manual IP assignment. This simplified network management and confirmed system scalability.

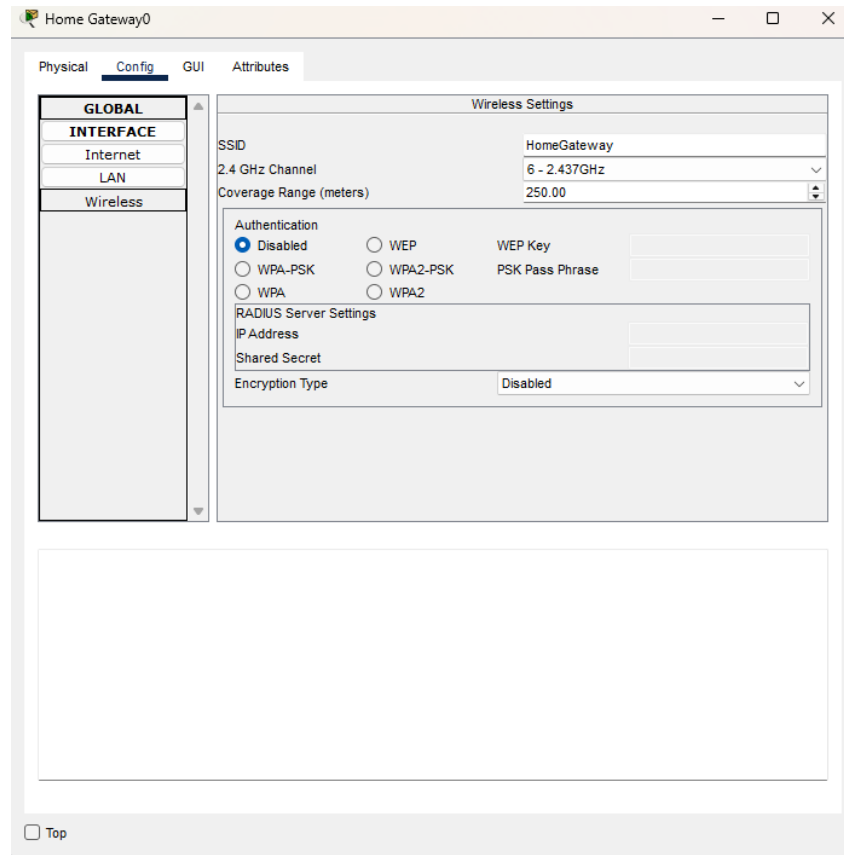


Figure 5: Gateway connectivity.

Each subsystem behaved as expected. The wild animal intrusion detection module triggered the siren upon motion detection, enhancing farm security. The soil moisture monitoring system activated sprinklers when the moisture level dropped below a threshold, showcasing water efficiency. The fire detection system, using a smoke sensor and microcontroller, activated both the siren and sprinkler during simulated fire conditions. Similarly, the solar panel module managed power flow through a microcontroller, switching to battery output when the charge exceeded the defined level.

The smart door system responded accurately to authorized RFID cards, unlocking the barricade and silencing the alarm, while unauthorized attempts kept the door locked and triggered the siren. All devices were ping-tested to ensure network connectivity, and the responses confirmed that communication via DHCP and the home gateway was stable and reliable.

Overall, the system validated the practical use of IoT and microcontroller-based logic in agriculture. It provided automated, real-time threat detection and environmental monitoring, making it a valuable prototype for sustainable and secure smart farming.

6 Advantages & Disadvantages

6.1 Advantages of IoT Devices in This Project

IoT devices in this project provide real-time monitoring and automation, reducing manual effort and improving efficiency. For example, the soil moisture sensor detects low moisture levels and automatically activates the sprinkler system, ensuring timely irrigation without human intervention. This kind of automation not only conserves resources but also increases reliability in farm operations.

6.2 Advantages

- **Automated IP Assignment:** DHCP allows devices to obtain IP addresses automatically, reducing the need for manual network configuration and minimizing human errors.
- **Scalability:** As more sensors or devices are added to the farm, the DHCP-based network can handle the expansion without additional setup, making it ideal for large-scale environments.
- **Centralized Power and Control:** Devices powered by solar panels and managed through microcontrollers ensure energy-efficient and coordinated automation of farm processes.
- **Ease of Maintenance:** With dynamic addressing and centralized logic, diagnosing and updating components becomes simpler and quicker.

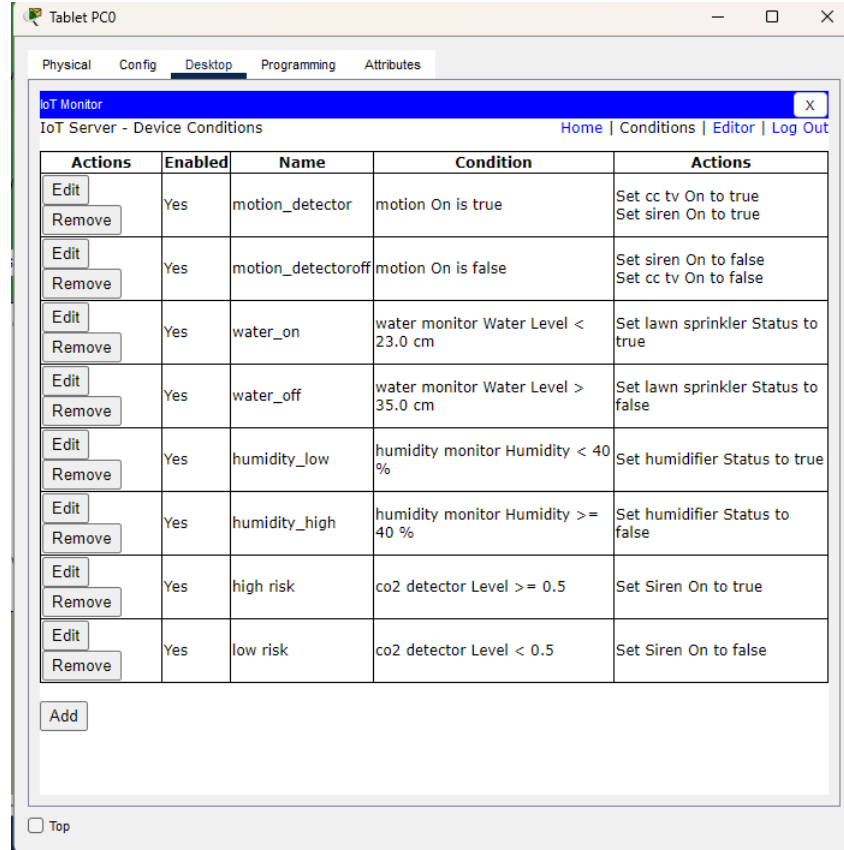


Figure 6: Predefined conditions of IoT devices on the design.

6.3 Disadvantages

- **Dependency on DHCP Server:** If the DHCP server or home gateway fails, devices may lose connectivity or fail to acquire IP addresses, disrupting farm operations.
- **Lease Expiration:** DHCP leases are temporary; if not renewed properly, devices might be reassigned or disconnected. While this can be adjusted by modifying lease durations, it still introduces a layer of dependency.
- **Network Reliability:** Since the system relies on wireless communication, signal interference or weak coverage may affect real-time responses or data transmission.

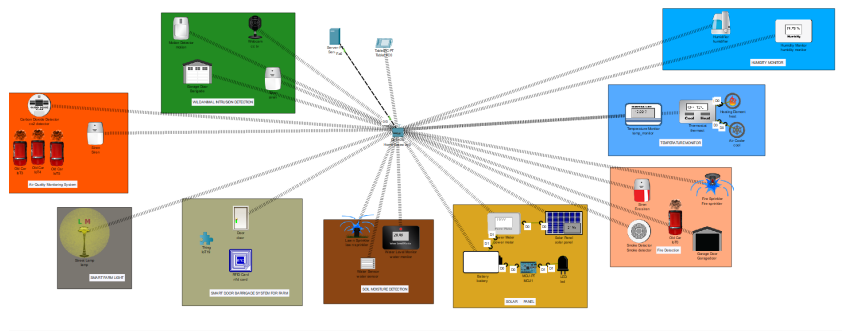


Figure 7: Logical view of the design.

7 Software Implementation

Example – Fire Detection Using Python:

```
from gpio import *
from time import *

def main():
    pinMode(0, INPUT)
    pinMode(1, OUT)
    while True:
        fire = digitalRead(0)
        if fire == 1023:
            customWrite(1, '1')
        else:
            customWrite(1, '0')

main()
```

Scripts are uploaded to microcontroller IDEs to handle logic for light, doors, sirens, etc.

8 Further Work

Future improvements could include integrating a Weather API to adjust irrigation based on real-time climate conditions, helping conserve water and improve crop health. The system can also be enhanced with SMS or email alerts to notify users of critical events like fire or unauthorized access. Additionally, applying predictive analytics to sensor data could help forecast plant needs and detect potential issues early, enabling smarter farm management.

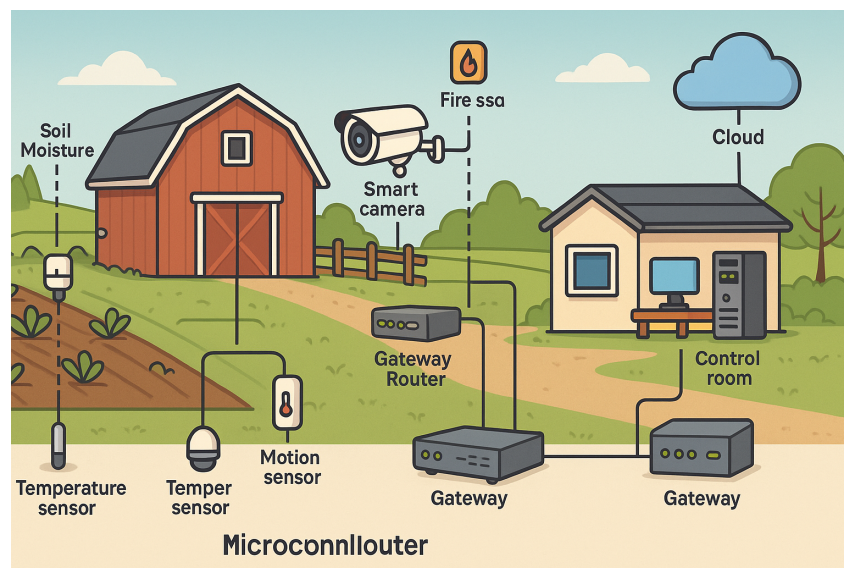


Figure 8: Physical view of the design.

9 Conclusion

With IoT and DHCP configuration, smart farming can be both efficient and flexible. DHCP enables scalable device addition without IP conflicts. Cisco Packet Tracer effectively models such intelligent farm systems, preparing for real-world deployment.

Conflicts of Interest

The authors declare no conflict of interest.

References

- [1] R.N. Rao and B. Sridhar, *IoT Based Smart Crop-Field Monitoring*, 2nd International Conference on Inventive Systems and Control (ICISC), 2018, pp. 478–483.
- [2] H. Gupta, V. Pareek, R. Mishra, and K. Lavanya, *Automated Precision Farming Using IoT*, International Research Journal of Advanced Research in Computer Engineering and Technology, vol. 6, 2017, p. 16.